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Exploring RGB and Video V-JEPA 2.1 Features for Aerial Depth Estimation and Semantic Scene Completion

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Abstract. Aerial 3D scene understanding is essential for autonomous UAV operations such as cargo delivery, search-and-rescue, and human-UAV collaboration. Camera-based perception is attractive for these settings given its light weight, low cost, availability, and compatibility with UAV size, weight, and power constraints. This motivates methods that can infer geometry, semantics, and free space from visual observations, making Monocular Depth Estimation (MDE) and Semantic Scene Completion (SSC) key components of UAV 3D perception that enable 3D scene understanding. Aerial imagery introduces altitude-dependent scale variation, oblique viewpoints, weak texture, and different occlusion patterns, demanding domain-specific solutions. Using OccuFly, a real-world aerial dataset with RGB images, metric depth maps, and semantic occupancy grids, we study whether V-JEPA 2.1 foundation features transfer to aerial MDE and SSC. Our ablations progress from V-JEPA 2.1 image-spatial & video-spatiotemporal feature handling, from linear probing to DPT (Dense Prediction Transformer)-style multiscale decoding, FiLM (Feature-wise Linear Modulation)-based altitude & intrinsics conditioning for MDE, followed by injecting conditioned depth-estimation priors, deploying a learnable UAV pose-aware multi-view stereo depth-estimation module, and transferring spatiotemporal features for SSC. Through this staged analysis, we separate the contributions of representation transfer, altitude conditioning, and depth-estimation variants to improving aerial SSC.

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Keywords: Aerial · 3D Scene Understanding · UAV Perception · Monocular Depth Estimation · Semantic Scene Completion · Semantic Occupancy Prediction · V-JEPA 2.1 · Video Foundation models · 2D-3D lifting · Pose-aware multi-view stereo depth · OccuFly

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1 Introduction

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Aerial 3D scene understanding is essential for autonomous UAV operations such as cargo delivery, search-and-rescue, and human-UAV collaboration. Compared with ground robots or autonomous vehicles, UAVs operate under strict payload, energy, and compute constraints, making lightweight RGB cameras attractive for perception. However, aerial imagery introduces viewpoint and scale shifts

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that differ substantially from indoor and automotive settings demanding domain specific solutions. Earlier UAV datasets such as UAVid focused mainly on 2D semantic segmentation from high-resolution oblique UAV video and highlighted scale variation, moving objects, and temporal consistency as key challenges [22]. UrbanScene3D broadened aerial research toward large-scale urban reconstruction and simulation, but it was not designed primarily for per-frame semantic scene completion [20]. OccuFly directly targets this gap by providing a real-world aerial SSC benchmark with RGB images, metric depth maps, semantic occupancy grids, camera metadata, multiple altitudes, and urban/industrial/rural scenes [9]. This makes OccuFly a suitable testbed for studying whether visual foundation features transfer to aerial metric depth estimation and semantic scene completion.

Monocular Depth Estimation (MDE) is underconstrained because a single RGB image does not uniquely determine metric depth; the model must infer depth from perspective, texture, object size, shadows, scene layout, and learned priors. Eigen et al. showed that coarse global prediction combined with local refinement can learn single-image depth, while also highlighting scale ambiguity as a central challenge [8]. MiDaS improved cross-dataset robustness by mixing heterogeneous depth datasets with compatible objectives [30]. DPT then replaced convolutional backbones with vision transformers for dense prediction, reassembling transformer tokens from multiple layers into image-like feature maps at different resolutions [29]. More recent models such as ZoeDepth and Depth Anything/Depth Anything V2 combine large-scale pretraining, relative-depth generalization, metric-depth adaptation, synthetic data, and pseudo-labeled real data to improve zero-shot and metric depth performance [4, 36, 37]. However, aerial MDE remains distinct because altitude, camera intrinsics, nadir/oblique view angle, and weak texture change the relationship between appearance and depth.

Semantic Scene Completion (SSC) jointly predicts volumetric occupancy and semantic labels, thereby combining geometry, free space, and semantic understanding in a single 3D representation. SSCNet introduced this formulation from a single depth image using a 3D convolutional architecture, showing that occupancy completion and semantic labeling are tightly coupled [33]. Later work moved toward RGB-only and camera-based SSC. MonoScene formulated monocular RGB SSC by projecting 2D image features into 3D and using 3D context priors [5]. VoxFormer used depth estimation to initialize sparse visible voxel queries before dense voxel completion, making depth a key bridge between 2D images and 3D semantic volumes [17]. TPVFormer represented 3D space using three orthogonal planes rather than a single BEV plane, while SurroundOcc and OpenOccupancy expanded occupancy prediction toward surrounding multi-camera autonomous-driving settings [13, 34, 35]. CGFormer further addressed depth ambiguity by using context-dependent voxel queries and geometry-aware lifting in 3D pixel space [40]. These works show that camera-based SSC depends strongly on how 2D features are lifted into 3D, but most are developed for indoor or automotive viewpoints rather than aerial UAV imagery.

Self-supervised visual foundation models aim to learn transferable representations without dense task labels.

In a posed MVS module, parallax becomes a measurable reprojection or feature-alignment error. While MDE relies heavily on learned priors, Multi-View Stereo (MVS) estimates depth from correspondence across calibrated views. Learning-based MVS methods such as MVSNet construct a plane-sweep cost volume by warping source-view features into a reference view over multiple depth hypotheses using known camera intrinsics and relative poses; the correct depth should produce better feature alignment and therefore lower matching cost [38]. R-MVSNet reduced the memory cost of full 3D cost-volume regularization by sequentially processing depth hypotheses, while CasMVSNet used a cascade strategy that narrows the depth range from coarse to fine stages [10, 39]. DeepVideoMVS extended this idea to posed video streams, propagating geometry over time with recurrent spatio-temporal fusion [7]. For aerial SSC, a pose-aware MVS frontend is attractive because neighboring UAV frames can provide motion parallax that is unavailable to a monocular depth head. Instead of using temporal video features only as generic sequence context, pose-aware MVS explicitly tests depth hypotheses through calibrated reprojection.

MAE showed that masked image reconstruction scales effectively for vision transformers [11]. I-JEPA introduced a non-generative joint-embedding predictive architecture that predicts target-block representations from context representations rather than reconstructing pixels [1]. V-JEPA extended this predictive latent-space objective to video by predicting missing spatio-temporal regions in representation space rather than pixel space [3]. V-JEPA 2 scaled the approach to internet-scale video and image data for understanding, prediction, and planning [2]. V-JEPA 2.1 is particularly relevant for dense prediction because it explicitly targets dense visual representation learning through dense predictive loss, deep self-supervision across intermediate encoder layers, and unified image/video tokenization [25]. These properties make V-JEPA 2.1 a strong candidate backbone for aerial MDE and SSC, but they do not automatically guarantee metric depth or volumetric occupancy accuracy; therefore, frozen probing, decoder ablations, metadata conditioning, and pose-aware MVS are necessary to test what geometry is actually encoded.

At the same time, modern visual foundation models such as V-JEPA 2.1 provide dense image/video representations that may encode spatial structure, semantics, and temporal continuity, but their transfer to aerial metric depth and semantic occupancy is not yet well understood. This work therefore studies V-JEPA 2.1 as a frozen image/video backbone for OccuFly, separating the effects of representation transfer, decoder structure, altitude/intrinsics conditioning, monocular depth priors, and explicit pose-aware multiview geometry.

Camera-based perception is attractive for these settings given its light weight, low cost, availability, and compatibility with UAV size, weight, and power constraints. This motivates methods that can infer geometry, semantics, and free space from visual observations, making Monocular Depth Estimation (MDE) and Semantic Scene Completion (SSC) key components of UAV 3D percep-

tion that enable 3D scene understanding. Aerial imagery introduces altitude-dependent scale variation, oblique viewpoints, weak texture, and different occlusion patterns, demanding domain-specific solutions. Using OccuFly, a real-world aerial dataset with RGB images, metric depth maps, and semantic occupancy grids, we study whether V-JEPA 2.1 foundation features transfer to aerial MDE and SSC. Our ablations progress from V-JEPA 2.1 image-spatial & video-spatiotemporal feature handling, from linear probing to DPT (Dense Prediction Transformer)-style multiscale decoding, FiLM (Feature-wise Linear Modulation)-based altitude & intrinsics conditioning for MDE, followed by injecting conditioned depth-estimation priors, deploying a learnable UAV pose-aware multi-view stereo depth-estimation module, and transferring spatiotemporal features for SSC. Through this staged analysis, we separate the contributions of representation transfer, altitude conditioning, and depth-estimation variants to improving aerial SSC.

Earlier UAV datasets such as UAVid focused mainly on 2D semantic segmentation from high-resolution oblique UAV video and highlighted scale variation, moving objects, and temporal consistency as key challenges [22]. UrbanScene3D broadened aerial research toward large-scale urban reconstruction and simulation, but it was not designed primarily for per-frame semantic scene completion [20]. OccuFly directly targets this gap by providing a real-world aerial SSC benchmark with RGB images, metric depth maps, semantic occupancy grids, camera metadata, multiple altitudes, and urban/industrial/rural scenes [9]. This makes OccuFly a suitable testbed for studying whether visual foundation features transfer to aerial metric depth estimation and semantic scene completion.

Aerial three-dimensional scene understanding is a central capability for unmanned aerial vehicles (UAVs) operating in navigation, mapping, infrastructure inspection, search-and-rescue, human collaboration, and environmental monitoring. In these settings, an autonomous platform must not only recognize visible image regions, but also infer metric geometry, free space, and the semantic occupancy of partially observed volumes. Such a representation is important for collision avoidance, viewpoint planning, landing-zone assessment, and interaction with objects and humans in cluttered outdoor environments. Camera-based perception is particularly attractive for UAVs because RGB cameras are lightweight, inexpensive, widely available, and compatible with strict size, weight, and power constraints. However, inferring dense 3D structure from monocular or short video observations remains substantially harder than recognition, since the image formation process collapses scale, depth, and occlusion into a two-dimensional signal. Aerial video offers an additional source of geometry through motion parallax, but exploiting it requires more than generic temporal feature aggregation: the system must relate target and source views through camera calibration, relative pose, and depth hypotheses.

Most progress in semantic scene completion (SSC) and camera-based 3D perception has been driven by indoor reconstruction and autonomous-driving benchmarks. Early SSC formulations used depth as input to jointly predict volumetric occupancy and semantics [33]. Later monocular methods, such as MonoScene [5],

moved the problem toward RGB-only 2D-to-3D lifting, while VoxFormer [17], TPVFormer [13], SurroundOcc [35], OpenOccupancy [34], CGFormer [40], and FoundationSSC [6] developed increasingly capable view-transformation, voxel-attention, and semantic-geometric fusion designs. These works demonstrate that explicit 3D representations are valuable for scene-level reasoning, but their primary evaluation domains remain road scenes with ground-level perspective, relatively constrained camera height, and characteristic automotive object layouts. Aerial viewpoints violate many of these assumptions: altitude changes the apparent scale of all objects, oblique views produce mixed top-down and side-view evidence, fine semantic boundaries can be weakly textured, and occlusions differ from those observed by ground vehicles. Consequently, strong methods trained or designed for terrestrial scenarios do not necessarily transfer to UAV perception without careful adaptation.

OccuFly [9] creates an opportunity to study this gap in a real-world aerial setting. It provides RGB images, metric depth maps, camera calibration, and semantic occupancy grids for aerial scenes captured at multiple altitudes and across urban, industrial, and rural environments. Unlike earlier UAV semantic segmentation datasets such as UAVid [22], which focus on 2D dense labeling, or aerial reconstruction resources such as UrbanScene3D [20], OccuFly directly supports the joint study of metric depth estimation and 3D semantic scene completion from aerial camera observations. This makes it possible to ask whether modern visual foundation representations provide transferable geometric and semantic priors for aerial 3D scene understanding, rather than only for image classification, 2D segmentation, or ground-level depth prediction. It also enables a more precise question: when multiple calibrated UAV views are available, does explicit multiview correspondence provide a stronger geometric prior for SSC than a monocular depth estimator alone?

In this work we investigate V-JEPA 2.1 [25] as a frozen foundation representation for aerial depth estimation and depth-aware semantic scene completion. V-JEPA 2.1 and its successors learn by predicting representations in latent space rather than reconstructing pixels [2, 3]. V-JEPA 2.1 further emphasizes dense predictive supervision, deep self-supervision at intermediate layers, and unified image/video tokenization, making it a natural candidate for dense 3D perception. The central question of this paper is not whether a large foundation model performs well on generic vision tasks, but whether its spatial and spatio-temporal features encode the aerial geometry needed for metric depth and volumetric semantic completion. We therefore keep the backbone frozen and vary the decoder, metadata conditioning, temporal aggregation, and geometric lifting mechanism.

We study this question along three axes. First, for monocular depth estimation, we compare frozen linear probes against structured DPT-style decoders [29]. The linear probes test how linearly accessible metric depth is from V-JEPA 2.1 features, while the DPT decoders test whether multiscale token reassembly and convolutional fusion can recover fine spatial detail from frozen transformer features. We also examine a Simple Feature Pyramid (SFP) adapter inspired by ViTDet [16], which converts a single final-layer feature map into a

pyramid suitable for dense prediction. Because aerial scale is strongly coupled to flight altitude and camera calibration, we further condition selected decoders with feature-wise linear modulation (FiLM) [27] using altitude and camera intrinsics. Second, we compare spatial RGB features with spatio-temporal video features. Aerial videos contain motion parallax and temporal continuity that may help resolve depth ambiguities, but such cues are not guaranteed to be preserved in frozen video representations. We therefore contrast image-only features, repeated-frame controls, shuffled temporal controls, spatio-temporal linear probes, and dense temporal attention adapters.

Third, for SSC, we compare monocular and multiview geometric priors inside depth-aware lifting architectures inspired by Lift-Splat-Shoot [28], CGFormer [40], and FoundationSSC [6]. The monocular V-JEPA 2.1/FoundationSSC branch uses a DPT-FiLM depth-probability frontend to lift semantic context into the OccuFly voxel grid. We further introduce a Pose-MVS-Cost-Volume FoundationSSC variant that replaces the monocular depth-probability frontend with a posed multiview stereo cost volume in the spirit of learned MVS cost-volume methods [38]. Given a target UAV view and neighboring calibrated source views, this branch matches target/source V-JEPA 2.1 video features across metric depth hypotheses, regularizes the resulting cost volume, and converts it into the depth distribution consumed by the FoundationSSC lifting module. We also evaluate a Pose-MVS-FiLM variant that applies altitude/intrinsics FiLM conditioning directly to the MVS depth-bin logits before the depth softmax, so that the dense depth map and lifting probabilities remain consistent.

This design separates two different uses of video. Dense temporal attention tests whether frozen V-JEPA 2.1 video tokens contain useful target-frame depth cues after nonlinear aggregation. In contrast, Pose-MVS tests an explicitly geometric hypothesis: calibrated source views should constrain depth through projective correspondence, even when monocular appearance cues are ambiguous. This distinction is important for aerial perception, where altitude variation, repeated textures, vegetation, roofs, and oblique viewing directions can make single-image depth uncertain while neighboring UAV frames may provide exploitable parallax.

Our contributions are as follows:

- We present a systematic study of frozen V-JEPA 2.1 spatial and spatio-temporal representations for aerial metric depth estimation on OccuFly.
- We compare representation probes of increasing structure: linear depth heads, DPT-style multiscale decoders, SFP-adapted final-layer decoders, and altitude/intrinsic conditioned FiLM variants.
- We analyze the role of temporal evidence through video-feature probes and temporal-attention adapters, including controls that separate true temporal context from repeated or shuffled frame inputs.
- We extend the investigation from depth to semantic scene completion by incorporating predicted depth and depth distributions as geometric priors in LSS-, CGFormer-, and FoundationSSC-inspired aerial SSC variants.

- We introduce Pose-MVS-Cost-Volume FoundationSSC, which replaces monocular depth-probability lifting with pose-aware multiview cost-volume depth estimation from target/source UAV views.
- We further evaluate Pose-MVS-FiLM, a metadata-conditioned MVS variant that modulates cost-volume depth logits with altitude and camera intrinsics while preserving probability–depth consistency for voxel lifting.
- We identify the practical domain gap between ground-level camera-centric 3D perception and aerial 3D scene understanding, providing a focused evaluation protocol for future foundation-model transfer studies in UAV perception.

2 Related Work

2.1 Aerial scene understanding and aerial 3D benchmarks

Aerial perception has traditionally been studied through 2D semantic segmentation, object detection, mapping, and reconstruction. UAVid [22] introduced high-resolution oblique UAV videos for semantic segmentation and highlighted challenges such as scale variation, moving objects, and temporal consistency. Other aerial resources, including UrbanScene3D [20], support large-scale urban reconstruction and simulation from aerial observations. These datasets are valuable for 2D and reconstruction-oriented aerial perception, but they do not directly provide the combination of RGB images, metric depth, camera calibration, and per-frame semantic occupancy grids needed to study aerial SSC.

OccuFly [9] addresses this limitation by introducing a real-world aerial benchmark for camera-based 3D semantic scene completion. It provides aerial RGB images, metric depth maps, semantic voxel grids, and camera metadata over multiple scenes, altitudes, environments, and seasons. The dataset is especially relevant for UAV perception because it captures the geometric ambiguity of low-altitude flight: object scale depends on altitude, many surfaces are seen from oblique or partially top-down viewpoints, and free-space reasoning must be performed in a 3D volume rather than only on the image plane. Our work uses OccuFly not merely as an evaluation dataset, but as a testbed for studying whether dense image/video foundation features and explicit multiview geometry transfer to aerial depth and semantic occupancy.

2.2 Monocular depth estimation and dense prediction decoders

Monocular depth estimation is inherently ambiguous because metric depth must be inferred from perspective, texture, object size, scene layout, and learned priors. Early deep approaches used coarse-to-fine convolutional networks and scale-aware objectives [8]. Later methods improved dense prediction through stronger encoders, multiscale decoders, and more diverse training data. MiDaS demonstrated that mixing heterogeneous datasets with compatible objectives improves cross-dataset robustness [31]. DPT replaced purely convolutional backbones with

vision transformers and reassembled intermediate tokens into multiscale image-like features for dense prediction [29]. ZoeDepth combined relative-depth generalization with metric-depth specialization [4], while Depth Anything and Depth Anything V2 scaled monocular depth training with large pseudo-labeled and synthetic data, producing robust general-purpose depth models [36, 37].

These models show that large-scale pretraining and dense decoders are crucial for depth, but aerial depth remains a distinct transfer problem. UAV images often lack the ground-level cues used by indoor and automotive models: the horizon may be absent, object size priors vary with altitude, repetitive roofs and vegetation create weak texture, and far-range depth errors can dominate voxel lifting. For this reason, our work does not treat a single depth model as a black-box solution. Instead, we compare lightweight linear probes with DPT-style decoders on frozen V-JEPA 2.1 features, evaluate final-layer SFP adaptation inspired by ViTDet [16], and condition selected decoders with altitude and camera intrinsics through FiLM [27]. This lets us separate three factors: the geometric information already present in the representation, the spatial reconstruction capacity of the decoder, and the usefulness of explicit aerial metadata.

2.3 Multiview stereo and cost-volume geometry

Monocular depth relies heavily on learned priors, whereas multiview stereo (MVS) estimates depth by matching observations across calibrated views. Learning-based MVS methods such as MVSNet build a plane-sweep cost volume by warping source-view features into a reference view over multiple depth hypotheses, regularize the cost volume with 3D convolutions, and infer depth from a probability distribution over planes [38]. Recurrent MVSNet reduces the memory footprint of high-resolution MVS by replacing full 3D cost-volume regularization with recurrent processing over depth hypotheses [39]. These methods are designed for depth reconstruction, but the underlying principle is directly relevant to aerial SSC: when camera poses are available, target/source correspondence can provide geometric evidence that is complementary to monocular appearance priors.

Our Pose-MVS variants adapt this cost-volume principle to frozen V-JEPA 2.1 video features and downstream semantic scene completion. Rather than producing only a target-view depth map, the MVS branch produces a categorical depth distribution aligned with the FoundationSSC lifting bins. The resulting distribution is used to place semantic context into the OccuFly voxel grid, making MVS a geometric frontend for SSC rather than a standalone reconstruction module. This differs from the dense temporal attention depth experiments: temporal attention learns how to aggregate video tokens, while Pose-MVS explicitly projects source views into the target camera using intrinsics, relative pose, and metric depth hypotheses. The Pose-MVS-FiLM variant further asks whether altitude and camera calibration, already useful for monocular DPT-FiLM depth, should also modulate multiview cost-volume logits before depth probabilities are formed.

2.4 Self-supervised visual foundation models

Self-supervised visual foundation models aim to learn transferable representations without task-specific labels. Masked autoencoders showed that high-ratio masked image reconstruction can scale vision transformers effectively [11]. I-JEPA introduced a non-generative joint-embedding predictive architecture for images, predicting target-block representations from context representations instead of reconstructing pixels [1]. DINOv2 demonstrated that carefully curated large-scale self-supervised training can yield robust all-purpose image features [26], and DINOv3 further emphasizes high-resolution dense features and broad transfer across natural and aerial imagery [32].

Video representation learning extends these ideas to temporal structure. V-JEPA 2.1 learns video representations by predicting masked spatio-temporal features in latent space and can be used as a frozen backbone with lightweight probes [3]. V-JEPA 2.1 2 scales this approach to internet-scale video and image data for understanding, prediction, and planning [2]. V-JEPA 2.1 further modifies the recipe toward dense image/video features through dense predictive loss, deep self-supervision, and multimodal tokenizers [25]. These properties make V-JEPA 2.1 attractive for aerial 3D perception, where one needs spatial layout, temporal consistency, and view-dependent geometry. However, strong generic dense features do not guarantee metric depth accuracy or reliable 3D lifting under aerial viewpoints. Our work therefore evaluates V-JEPA 2.1 through explicit downstream depth, cost-volume, and SSC probes rather than only through representation visualization or global recognition metrics.

2.5 2D-to-3D lifting and camera-based semantic scene completion

Semantic scene completion jointly estimates occupied and free voxels together with semantic labels. SSCNet introduced the task using single-view depth input and 3D convolutional reasoning [33]. MonoScene extended SSC to monocular RGB by projecting 2D features into 3D and using 3D context priors [5]. Camera-based outdoor SSC then evolved toward sparse voxel queries, deformable attention, and explicit view transformation. VoxFormer uses estimated depth to initialize sparse visible voxel queries and then densifies them with transformer reasoning [17]. TPVFormer represents 3D space through three orthogonal planes, improving over a single BEV plane for semantic occupancy prediction [13]. SurroundOcc and OpenOccupancy expanded the problem toward surrounding multi-camera occupancy perception in autonomous driving [34, 35].

A recurring theme in these works is the importance of depth-aware lifting. Lift-Splat-Shoot first formalized a camera-to-BEV representation by predicting a depth distribution for each image feature and splatting frustum features into a common spatial grid [28]. BEVDepth later showed that explicit depth supervision improves camera-based BEV 3D detection [18]. For SSC, CGFormer addresses depth ambiguity by generating context-dependent voxel queries and extending deformable cross-attention from 2D image space to 3D pixel space; it also combines voxel and tri-perspective-view representations to capture local

and global 3D context [40]. FoundationSSC argues that semantic and geometric priors should be decoupled at both source and pathway levels, using foundation features and geometric depth distributions before hybrid 3D fusion [6].

Our SSC design follows this line of work but differs in domain and experimental goal. Existing methods are primarily optimized for indoor or autonomous-driving scenes, whereas we target aerial observations from OccuFly. We use predicted monocular depth and depth probabilities as geometric priors for voxel lifting, while V-JEPA 2.1 features provide semantic context. We then add Pose-MVS-Cost-Volume FoundationSSC to test whether calibrated target/source matching improves the depth distribution used for lifting, and Pose-MVS-FiLM to test whether altitude and intrinsics should condition that multiview distribution. This allows us to compare three sources of geometry under a shared aerial SSC protocol: monocular depth priors, video-feature temporal aggregation, and explicit posed multiview correspondence.

2.6 Position of this work

The closest work to ours is OccuFly [9], which establishes the aerial SSC benchmark and reports baseline limitations of existing methods. Our work is complementary: rather than proposing a new dataset, we use OccuFly to analyze the transfer behavior of a recent video/image foundation model for coupled aerial 3D tasks. Compared with generic monocular depth models such as Depth Anything V2 [37], we focus on what can be recovered from frozen V-JEPA 2.1 features and how decoder structure, metadata conditioning, temporal context, and posed multiview matching affect metric depth. Compared with camera-based SSC methods such as MonoScene, VoxFormer, CGFormer, and FoundationSSC [5, 6, 17, 40], we explicitly study aerial depth priors and foundation features under UAV-specific viewpoint and scale shifts. Compared with standard MVS methods such as MVSNet and R-MVSNet [38, 39], our goal is not standalone depth reconstruction but semantic occupancy: the cost volume is used to produce the depth-probability tensor that drives FoundationSSC-style voxel lifting. The resulting study connects four previously separate threads—aerial benchmarks, dense foundation representations, multiview cost-volume geometry, and depth-aware semantic scene completion—into a single evaluation of aerial 3D scene understanding.

3 Method

This section describes our evaluation framework for transferring frozen V-JEPA 2.1 image-spatial and video-spatiotemporal features to aerial metric depth estimation and semantic scene completion (SSC). The method starting is intentionally organized as a sequence of probes followed by progressively more structured decoders for evaluating dense feature transfer in both rgb and video variants of V-JEPA 2.1 for aerial MDE, with aim to select the best performer and compare with OccuFly dataset Finetuned DAV2 MDE. Decoupled Geometric and Image

context layers style SSC paradigm is chosen, specifically CGFormer and FoundationSSC inspired variants as they structurally allow for controlled evaluation on geometric and image context feature contribution for SSC.

We evaluate if and how far previously selected learned MDE heads, a learnable pose aware Multi View Stereo Depth Estimation head and a learnable image context feature represenattion head with their backbone as frozen V-JEPA 2.1, benifit in improving aerial SSC. Knowing that existing metric MDE models are not generalisable we leverage the fact that UAV pose/altitude and onboard camera intrinsics are usually known or estimated using auxilary systems and evaluate benifits of Feature wise learning based modulation (FiLM) for aerial MDE and indirectly SSC.

We mathematically formulate the train/eval/test data source, problem statement for MDE and SSC. Followed by description of the adapter/neck needed to make V-JEPA 2.1 features usable by the downstream heads. Then progressively dive deeper into considered MDE heads and SSC variants related sourcing, compatability details followed by answering a questionair that answers what componet or componet combination has helped to benifit aerial SSC?? Which has also motivated the direction of flow of experiments. Fig ?? outlines the high-level overview of the evaluation frame work, metrics and experiments conducted. Later the same structure will be used to report results obtained.

3.1 Problem Formulation and Notation

We use Occufly dataset [9] for all experiments. The per sample data details used from the dataset is mathematically formulated as follows.

$$s_t = (I_t, D_t, M_t, Y_t, K_t, h_t, T_t^{w \rightarrow c}), \quad (1)$$

where $I_t \in \mathbb{R}^{3 \times H \times W}$ is an RGB frame, D_t is a metric depth map, $M_t \in \{0, 1\}^{H \times W}$ is the valid-depth mask, Y_t is the semantic occupancy grid, K_t is the camera intrinsic matrix, h_t is the UAV altitude, and $T_t^{w \rightarrow c} \in \mathbb{R}^{4 \times 4}$ is the world-to-camera pose. The leading dimension 3 in I_t denotes RGB color channels, $H \times W$ denotes the image resolution before the V-JEPA 2.1 resize, and the 4×4 pose matrix is a homogeneous rigid transform.

The monocular depth, multiview depth, and SSC task functions are

$$\begin{aligned} f_D^{\theta_D} &: (I_t, K_t, h_t) \mapsto \hat{D}_t, \\ f_{D, \text{MVS}}^{\theta_M} &: (\mathcal{V}_t, h_t) \mapsto (\hat{D}_t^{\text{MVS}}, P_t^{D, \text{MVS}}), \\ f_S^{\theta_S} &: (\Phi_t, K_t, h_t, \hat{D}_t, P_t^D) \mapsto \hat{Y}_t. \end{aligned} \quad (2)$$

Here $f_D^{\theta_D}$ denotes the monocular depth-estimation model with trainable depth parameters θ_D , and $f_{D, \text{MVS}}^{\theta_M}$ denotes the learnable pose-aware multiview stereo depth estimator with parameters θ_M . The multiview input is

$$\mathcal{V}_t = \{(I_{t_i}, K_{t_i}, T_{t_i}^{w \rightarrow c})\}_{i=0}^{N_s}, \quad t_0 = t, \quad (3)$$

Here t is the target time index for which depth and semantic occupancy are predicted. The notation $t_0 = t$ means that the first element of $\mathcal{V}_t$ is the target, or reference, view. The remaining elements $t_1, \dots, t_{N_s}$ are calibrated source views used for multiview matching against the target view. We consider only two views for this study.

In the implementation, $f_{D,MVS}^{\theta_M}$ is realized by source-specific V-JEPA 2.1 video features, pose-aware projective matching, and a trainable cost-volume regularizer. At this level, we write it generally as a mapping from posed views to a target-frame dense depth estimate $\hat{D}_t^{MVS}$ and depth probability tensor $P_t^{D,MVS}$.

The function $f_S^{\theta_S}$ denotes the semantic scene completion model with trainable SSC parameters θ_S . In our frozen-backbone setting, the V-JEPA 2.1 encoder weights are fixed, while θ_D , θ_M , and θ_S are either learnable or frozen based on the variant. The term Φ_t denotes the visual context representation extracted from the target observation. The camera intrinsics K_t and altitude h_t provide geometric metadata for projection and scale conditioning. The dense depth prediction $\hat{D}_t$ gives the expected metric depth at each image location, while P_t^D is a depth probability tensor used for lifting image features along camera rays. The output $\hat{Y}_t$ is the predicted semantic occupancy grid.

3.2 Frozen V-JEPA 2.1 Feature Extraction

V-JEPA 2.1 is used in two input modes: an RGB/image mode and a video mode. Let E_θ denote the frozen V-JEPA 2.1 encoder [25]. For the ViT-Gigantic checkpoint used in the experiments, the feature channel dimension is $C = 1664$, which is why every spatial token is represented by a 1664-dimensional descriptor. The model has 48 transformer blocks, indexed from 0 to 47, and a 384×384 input with patch size 16 producing a 24×24 spatial grid, $384/16 = 24$.

RGB mode: A target image I_t is encoded into one or more spatial feature maps. We use either the final layer alone or four intermediate layers:

$$\mathcal{L}_G = \{11, 23, 37, 47\}, \quad F_t^{(l)} = E_\theta^{(l)}(I_t) \in \mathbb{R}^{C \times 24 \times 24}, \quad l \in \mathcal{L}_G. \quad (4)$$

Here, $\mathcal{L}_G$ lists the selected ViT-G layers, l indexes one of those layers, and $F_t^{(l)}(:, u, v)$ is the 1664-dimensional RGB descriptor at spatial grid cell (u, v) . For image-mode depth prediction, the frozen encoder returns image-grid features without a temporal tubelet axis. In particular, the final-layer RGB feature map has the form

$$F_t^{(47)} \in \mathbb{R}^{1664 \times 24 \times 24}. \quad (5)$$

A dense linear probe applies a lightweight batch-normalized 1×1 prediction head to the final frozen feature map, following standard frozen-feature linear probing practice for dense ViT representations [23, 25, 32]. The DPT variant instead consumes selected intermediate feature maps and reassembles them into a multiscale hierarchy following DPT [29].

Video mode: We form a four-frame clip centered at target frame t :

$$X_t = [I_{t+\tau_1}, I_{t+\tau_2}, I_{t+\tau_3}, I_{t+\tau_4}] \in \mathbb{R}^{3 \times 4 \times 384 \times 384}, \quad (6)$$

$$[\tau_1, \tau_2, \tau_3, \tau_4] = [-1, 0, 0, +1].$$

The duplicated zero offset places the target frame twice in the clip. This gives a target-centered local temporal window while keeping the clip length divisible by the V-JEPA 2.1 tubelet size.

For V-JEPA 2.1 ViT-Gigantic, the patch size is 16, the video tubelet size is 2, and the feature dimension is $C = 1664$. Therefore, a 384×384 input produces a 24×24 spatial grid, and a T -frame video clip produces $T' = T/2$ temporal tubelet positions:

$$G_t^{(l)} = E_\theta^{(l)}(X_t) \in \mathbb{R}^{1664 \times T' \times 24 \times 24}. \quad (7)$$

For the four-frame clips used here, $T = 4$ and hence $T' = 2$. Video-mode features contain both spatial locations and temporal tubelet tokens, so we use lightweight adapters to expose them to dense depth heads.

We tested two types of adapters for video features: The first type being *linear probe for video mode* that uses only the final V-JEPA feature tensor $G_t^{(47)}$. To keep the probe linear while retaining the tubelet information, the temporal tubelet axis is folded into the channel axis before applying the same 1×1 depth-bin head used by the image linear probe:

$$G_t^{(47)} \in \mathbb{R}^{1664 \times T' \times 24 \times 24} \mapsto \tilde{F}_t \in \mathbb{R}^{(1664T') \times 24 \times 24}, \quad (8)$$

$$\tilde{F}_t[t'C + c, u, v] = G_t^{(47)}[c, t', u, v], \quad 0 \leq c < 1664, \quad 0 \leq t' < T'.$$

For $T = 4$, this gives

$$\tilde{F}_t \in \mathbb{R}^{3328 \times 24 \times 24}. \quad (9)$$

This probe has no spatial decoder, temporal attention, recurrent module, or backbone finetuning. It is used as a diagnostic for whether depth-relevant temporal information is linearly accessible from the frozen final-layer video features.

The second adapter is the *Dense Temporal Attention* (DTA) adapter. DTA is inspired by the attentive probing heads used for V-JEPA 2.1 video evaluation, where learnable-query cross-attention pools frozen video tokens for classification [2, 24, 25]. Instead of pooling all spatio-temporal tokens into one global classification vector, DTA applies the same learned-query attention idea locally at every spatial cell. For each selected layer $l \in \{11, 23, 37, 47\}$, the temporal descriptors at location (u, v) are

$$S_{t,u,v}^{(l)} = \left[G_t^{(l)}[:, 1, u, v], \dots, G_t^{(l)}[:, T', u, v] \right] \in \mathbb{R}^{T' \times 1664}. \quad (10)$$

The adapter A_l attends over this local tubelet sequence and returns one image-grid feature vector at the same spatial location:

$$A_l(G_t^{(l)}) = \bar{F}_t^{(l)} \in \mathbb{R}^{1664 \times 24 \times 24}. \quad (11)$$

At a single spatial location, the operation can be summarized as

$$\begin{aligned}\mu_t^{(l)}[:, u, v] &= \frac{1}{T'} \sum_{\tau=1}^{T'} G_t^{(l)}[:, \tau, u, v], \\ \bar{F}_t^{(l)}[:, u, v] &= \mu_t^{(l)}[:, u, v] + \alpha_l \left(\text{DTA}_l(S_{t,u,v}^{(l)}) - \mu_t^{(l)}[:, u, v] \right),\end{aligned}\tag{12}$$

where $\text{DTA}_l(\cdot)$ denotes the bottleneck projection, learned-query multi-head temporal attention, MLP refinement, and expansion back to 1664 channels. The residual gate α_l is initialized near temporal mean pooling, so training can gradually introduce attention-based temporal corrections.

The resulting DTA maps $\{\bar{F}_t^{(l)}\}_{l \in \{11, 23, 37, 47\}}$ have the same $1664 \times 24 \times 24$ image-grid form as RGB V-JEPA feature maps. They can therefore be passed to the same dense downstream machinery used in image mode, including DPT-style decoding for depth prediction, F2D feature prediction, and pose aware multiview cost-volume construction.

3.3 Depth Estimation

Depth estimation in this study follows the monocular mapping $f_D^{\theta_D}$ and the pose-aware multiview stereo mapping $f_{D, \text{MVS}}^{\theta_M}$ in Eq. (2). The monocular heads use a target image, camera intrinsics, and altitude metadata, while the MVS heads use calibrated target/source views to estimate target-frame depth. In the variants considered here, each depth head predicts ray-wise logits over metric depth bins and converts them to a dense metric depth estimate:

$$p_{k,u,v} = \text{softmax}_k(z_{k,u,v}), \quad \hat{D}_{u,v} = \sum_{k=1}^{K_D} p_{k,u,v} d_k,\tag{13}$$

where $K_D = 256$ and d_k are metric depth-bin centers. Thus the depth output is represented consistently as a categorical depth probability tensor and its dense metric estimate. In the monocular heads the logits z are produced by a linear probe or DPT-style decoder; in the pose-aware MVS heads they are produced by a regularized projective cost volume before the same softmax-based depth-estimation conversion.

All variants in Table 1 are supervised with the OccuFly target-frame metric depth map D_t and valid-pixel mask Ω_D using the scale-invariant loss in Eq. (25); for Pose-MVS-FiLM, the same supervision is applied to the target-frame MVS estimate $\hat{D}_t^{\text{MVS}}$. The table summarizes the feature flow, representative tensor shapes, and learnable components; light-pink boxes mark the parts updated during training, while unmarked backbone features remain frozen.

Linear Probe The linear probe is the simplest transfer test in the depth study. It keeps the V-JEPA 2.1 backbone frozen and uses only the learnable head marked in Table 1, following standard frozen-feature dense probing practice and

Input split	OccuFly train/validation/test samples
↓	image (I_t, K_t, h_t) , video clip (X_t, K_t, h_t) , or posed views $\mathcal{V}_t$
Frozen backbone	V-JEPA 2.1 features $F_t^{(l)}$ or $G_t^{(l)}$
↓	LP / SFP-DPT / DPT-FiLM / Video-FiLM-DPT / Pose-MVS-FiLM
Trainable head	depth-bin logits
↓	
Output	depth probabilities P_t^D and estimated depth $\hat{D}_t$ or MVS counterparts $(P_t^{D,MVS}, \hat{D}_t^{MVS})$

Fig. 1: Common depth-estimation flow. The V-JEPA 2.1 backbone is frozen, while the depth head is trained on the official OccuFly train/validation/test split.

the DINOv3 open-source dense-depth linear head used as implementation reference for the RGB case [23, 32]. Its purpose is diagnostic: if a linear head can recover metric depth, then the frozen representation already exposes monocular depth estimation (MDE) cues in a directly transferable form.

The same diagnostic is applied in RGB and video mode, as summarized in Table 1. In RGB mode, the probe is applied to the final image feature map $F_t^{(47)}$. In video mode, the final video tensor $G_t^{(47)}$ is converted to the video linear-probe feature $\tilde{F}_t$ by folding the temporal tubelet axis into the channel axis before the same type of location-wise head is applied. This video probe therefore tests whether temporal features from V-JEPA 2.1 contain depth cues that are accessible without a nonlinear temporal decoder.

DPT Depth Decoders DPT-style decoders are the structured alternatives to the linear probes. They follow the Dense Prediction Transformer principle of reassembling transformer features into image-like maps and fusing them with convolutional refinement for dense prediction [29]. Table 1 gives the variant-level data flow, representative shapes, and learnable blocks; the equations below define only the shared DPT intermediates used by the SFP and FiLM variants.

For the multilayer DPT path, selected V-JEPA 2.1 layers $\mathcal{L}_G = \{11, 23, 37, 47\}$ are reassembled and projected to a common decoder width $C_D = 256$:

$$R_l = \rho_l \left(F_t^{(l)} \right), \quad R_l \in \mathbb{R}^{C_D \times H_l \times W_l}, \quad l \in \mathcal{L}_G, \quad (14)$$

where ρ_l denotes the DPT reassembly and projection block. These maps are either fused directly by the multilayer DPT decoder or passed through FiLM conditioning before fusion.

SFP-DPT. The Simple Feature Pyramid (SFP) variant tests the final-layer-feature hypothesis for V-JEPA 2.1 transfer. V-JEPA 2.1 reports strong dense features with deep supervision [25], so a plausible question is whether the final ViT-G map already carries enough earlier-layer information for depth once it

Table 1: Simplified component flow for the depth-head variants. Light-pink boxes denote the learnable components updated during training; unmarked V-JEPA 2.1 feature extraction remains frozen.

Variant	Data flow with representative shapes
RGB Probe	Linear $I_t \rightarrow F_t^{(47)}[1664, 24, 24] \rightarrow \text{BN} + 1 \times 1 \rightarrow Z_t[256, 24, 24] \rightarrow (\hat{D}_t, P_t^D)$.
Video Probe	Linear $X_t \rightarrow G_t^{(47)}[1664, T', 24, 24] \rightarrow \tilde{F}_t[(1664T'), 24, 24] \rightarrow \text{BN} + 1 \times 1 \rightarrow Z_t[256, 24, 24] \rightarrow (\hat{D}_t, P_t^D)$.
SFP-DPT	$I_t \rightarrow F_t^{(47)}[1664, 24, 24] \rightarrow \text{SFP}\{S_4, S_2, S_1, S_{1/2}\} \rightarrow \text{DPT} \rightarrow (\hat{D}_t, P_t^D)$, $S_s \in \{[256, 96, 96], [256, 48, 48], [256, 24, 24], [256, 12, 12]\}$.
Multilayer DPT	$I_t \rightarrow \{F_t^{(11)}, F_t^{(23)}, F_t^{(37)}, F_t^{(47)}\} \rightarrow \text{DPT reassembly/fusion} \rightarrow (\hat{D}_t, P_t^D)$.
FiLM multilayer DPT	$I_t, K_t, h_t \rightarrow \{R_t\}, e_t \rightarrow \text{FiLM}(\{R_t\}, e_t) \rightarrow \text{DPT fusion} \rightarrow (\hat{D}_t, P_t^D)$.
Video multilayer DPT	FiLM $X_t, K_t, h_t \rightarrow \{G_t^{(l)}[1664, T', 24, 24]\} \rightarrow \text{DTA} \rightarrow \{\tilde{F}_t^{(l)}[1664, 24, 24]\}, e_t \rightarrow \text{FiLM DPT} \rightarrow (\hat{D}_t, P_t^D)$.
Pose-MVS-FiLM	$\mathcal{V}_t, h_t \rightarrow G_i^{(l)}[1664, T', 24, 24] \rightarrow \text{DTA} \rightarrow \text{pose cost volume} \rightarrow \text{FiLM logits} \rightarrow (\hat{D}_t^{\text{MVS}}, P_t^{D, \text{MVS}})$.

is converted into a multiscale pyramid. To test this, the SFP-DPT branch uses only the final feature map $F_t^{(47)}$ and constructs the multiscale maps listed in Table 1. The pyramid design is inspired by the Simple Feature Pyramid used with plain ViT backbones in ViTDet [16]; we cite it as architectural inspiration rather than as an external implementation dependency.

$$\text{SFP}(F_t^{(47)}) = \{S_4, S_2, S_1, S_{1/2}\}, \quad S_s \in \mathbb{R}^{256 \times (24s) \times (24s)}, \quad s \in \{4, 2, 1, 1/2\}. \quad (15)$$

The resulting pyramid is decoded by the DPT-style depth head marked in Table 1 into 256 depth-bin logits. This variant separates the value of explicit multiscale decoding from the value of using multiple intermediate V-JEPA 2.1 layers.

FiLM-Conditioned DPT for Aerial Depth. The FiLM-conditioned DPT variants test whether altitude and camera calibration should modulate the depth decoder in aerial scenes. This is relevant because UAV altitude and intrinsics affect the mapping from pixels to metric rays, while the frozen V-JEPA 2.1 representation itself is not updated. The flow and tensor shapes for RGB and video FiLM-DPT variants are listed in Table 1; here we define the metadata conditioning applied to the DPT features.

The conditioning input is the normalized metadata vector

$$m_t = \left[\frac{h_t - 40}{10}, \frac{f_x}{W}, \frac{f_y}{H}, \frac{c_x}{W} - 0.5, \frac{c_y}{H} - 0.5 \right] \in \mathbb{R}^5, \quad (16)$$

where h_t is UAV altitude and (f_x, f_y, c_x, c_y) are camera intrinsics expressed relative to the image size used by the branch. A small MLP maps m_t to an embedding $e_t \in \mathbb{R}^{128}$. For each DPT level, learned affine projections produce a scale and shift that modulate the reassembled feature map:

$$\tilde{R}_l = R_l \odot (1 + \gamma_l(e_t)) + \beta_l(e_t), \quad (17)$$

where $\gamma_l(e_t)$ and $\beta_l(e_t)$ have the channel dimension of R_l and are broadcast over spatial locations. Thus FiLM does not change the spatial resolution listed in Table 1; it applies a learned, metadata-dependent channel-wise scale and shift before DPT fusion. The FiLM projections are initialized so that $\gamma_l \approx 0$ and $\beta_l \approx 0$ at the start of training, making Eq. (17) initially reduce to the unconditioned DPT path. The FiLM-modulated maps $\{\tilde{R}_l\}$ are then fused by DPT refinement blocks and decoded into 256 depth-bin logits.

For video FiLM-DPT, the DPT inputs are first converted from tubelet-aware video features to target-frame feature maps using Dense Temporal Attention (DTA), as shown in Table 1. At each selected layer and spatial site, DTA attends over the temporal tubelet descriptors and returns a target-frame map:

$$\bar{F}_t^{(l)}(:, u, v) = \mu_t^{(l)}(:, u, v) + \alpha \left(\text{Expand}(\text{MLP}(\text{MHA}(q, K, V))) - \mu_t^{(l)}(:, u, v) \right), \quad (18)$$

where $\mu_t^{(l)} = \text{mean}_\tau G_t^{(l)}(:, \tau, :, :)$ and $\alpha = \sigma(a)$ is initialized to approximately 0.1. The resulting maps $\{\bar{F}_t^{(l)}\}_{l \in \mathcal{L}_G}$ are passed to the same FiLM-conditioned DPT decoder as the RGB multilayer variant.

Pose-MVS-FiLM Pose-MVS-FiLM is the pose-aware multiview depth branch in Table 1. It keeps the V-JEPA 2.1 backbone frozen and uses calibrated target/source views to construct a depth-indexed cost volume, following the reference-view matching principle used in learned MVS cost-volume methods such as MVS-Net [38]. Let $\mathcal{V}_t = \{(I_i, K_i, T_{t \rightarrow i})\}_{i \in \mathcal{S}_t}$ denote the calibrated source views for target frame t . For target pixel $\mathbf{u} = (u, v, 1)^\top$ and metric depth-bin center d_k , the target-frame 3D point and source-view projection are

$$\mathbf{x}_{t,k,u,v} = d_k K_t^{-1} \mathbf{u}, \quad \mathbf{q}_{i,k,u,v} = \pi \left(K_i T_{t \rightarrow i} \begin{bmatrix} \mathbf{x}_{t,k,u,v} \\ 1 \end{bmatrix} \right), \quad (19)$$

where $\pi(\cdot)$ is perspective projection. Let A_t and A_i denote target/source feature maps after the video-feature adaptation shown in Table 1. Source features are sampled at the projected locations, compared with the target feature, and aggregated over source views into a depth-indexed cost volume,

$$\tilde{A}_{i,k,u,v} = \text{sample}(A_i, \mathbf{q}_{i,k,u,v}), \quad C_{t,k,u,v} = \text{Agg}_{i \in \mathcal{S}_t} \text{corr} \left(A_t(:, u, v), \tilde{A}_{i,k,u,v} \right). \quad (20)$$

Table 2: Depth-estimation variants and the transfer question isolated by each design. Learnable components are shown in Table 1.

Variant	Purpose
Linear probe	Tests whether metric aerial depth is linearly accessible from frozen V-JEPA 2.1 representations.
SFP-DPT	Tests whether the final V-JEPA 2.1 map can be converted into an effective dense pyramid without using intermediate layers.
Multilayer DPT	Tests whether selected intermediate layers improve dense metric depth over final-layer probing.
FiLM-DPT	Tests whether altitude/intrinsics conditioning improves RGB or video depth decoding.
Pose-MVS-FiLM	Tests whether calibrated multiview matching and meta-data conditioning improve the geometric prior.

The cost volume is regularized by a 3D CNN and mapped to FiLM-conditioned target-frame depth logits,

$$Z_t^{\text{MVS}} = \phi_{\text{logit}}(\psi_{3\text{D}}(C_t), e_t),$$

$$P_t^{D,\text{MVS}} = \text{softmax}_k Z_t^{\text{MVS}}, \quad \hat{D}_{t,u,v}^{\text{MVS}} = \sum_{k=1}^{K_D} P_{t,k,u,v}^{D,\text{MVS}} d_k. \quad (21)$$

This branch uses the same $K_D = 256$ metric depth-bin centers and target-frame depth supervision as Eq. (13). CVP-MVSNet and CasMVSNet are treated as related cascaded MVS literature where cited elsewhere; the implementation described here is a single cost-volume branch rather than a cascade of those methods.

Table 2 summarizes the depth variants and the question each variant is designed to answer. The learnable components for these variants are marked in Table 1, so this table focuses only on experimental motivation. The experiments form a controlled sequence from minimal representation probing to structured spatio-temporal decoding.

Altitude-Interpolation Depth Evaluation The standard OccuFly dataset split uses scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing, with altitudes 30 m, 40 m, and 50 m. We also include an altitude-interpolation protocol in which training uses 30 m and 50 m samples from scenes 01–08, validation uses 40 m samples from scenes 01–04, and testing uses 40 m samples from scenes 05–08:

$$\mathcal{D}_{\text{train}} = \{s_t : h_t \in \{30, 50\}, \text{scene}(t) \in \{01, \dots, 08\}\}, \quad (22)$$

$$\mathcal{D}_{\text{val}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{01, 02, 03, 04\}\}, \quad (23)$$

$$\mathcal{D}_{\text{test}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{05, 06, 07, 08\}\}. \quad (24)$$

This split is a scale-generalization stress test: it reports model depth prediction errors at the unseen middle altitude. It is especially relevant for FiLM-conditioned depth heads, because their metadata pathway is intended to use UAV altitude at inference time; performance at 40 m after training on 30 m and 50 m tests whether that conditioning supports interpolation across altitude rather than only fitting observed flight heights.

Losses The resulting depth map is resized to the supervision resolution and trained with a valid-pixel scale-invariant logarithmic loss. Let Ω_D be the set of valid supervised pixels after resizing, $N = |\Omega_D|$, and $i \in \Omega_D$ index one valid pixel. For predicted depth $\hat{D}_i$, ground-truth OccuFly depth D_i , and log-depth residual $g_i = \log \hat{D}_i - \log D_i$, the loss is

$$\mathcal{L}_{\text{sig}} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} g_i^2 - \lambda_{\text{si}} \left(\frac{1}{N} \sum_{i \in \Omega_D} g_i \right)^2} + \epsilon, \quad (25)$$

where λ_{si} controls the scale-invariant correction term and ϵ is a small positive constant for numerical stability. The term inside the square root is the scale-invariant log-depth objective of Eigen et al. [8] when $\lambda_{\text{si}} = 0.5$; the square root and ϵ give the numerically stabilized SiLog-style form used here. We use this loss because aerial metric depth contains unavoidable global-scale ambiguity, while the altitude/intrinsics conditioning experiments test whether that ambiguity can be reduced by physically meaningful metadata.

Evaluation Metrics Depth models are evaluated only over valid pixels. The manuscript uses root mean squared error (RMSE), absolute relative error (AbsRel), and log-RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} (\hat{D}_i - D_i)^2}, \quad \text{AbsRel} = \frac{1}{N} \sum_{i \in \Omega_D} \frac{|\hat{D}_i - D_i|}{D_i}, \quad (26)$$

$$\text{LogRMSE} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} (\log \hat{D}_i - \log D_i)^2}. \quad (27)$$

For video-depth runs, the evaluation script also supports MAE, squared-relative error, and threshold accuracies $\delta_1, \delta_2, \delta_3$ when enabled.

3.4 Semantic Scene Completion

Semantic scene completion (SSC) is evaluated after depth estimation because the depth branches provide geometry priors for downstream 3D semantic occupancy. The SSC experiments test whether those priors, together with the corresponding image-context or F2D features, transfer into a 3D completion task. The purpose is not to introduce a completely new SSC architecture. Instead, we use published

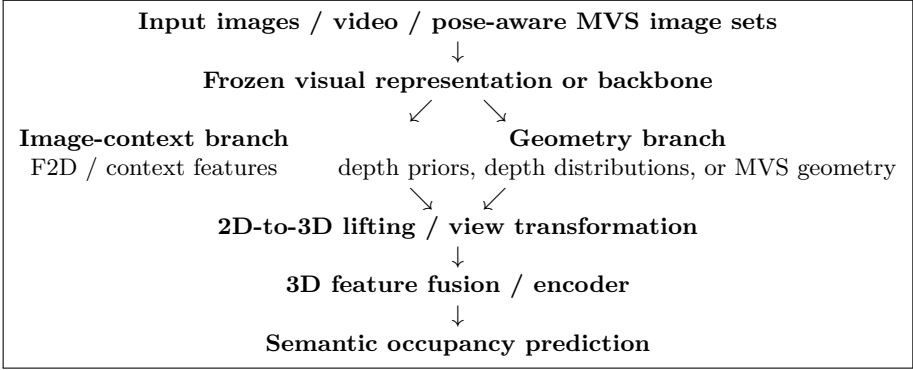


Fig. 2: Conceptual SSC evaluation flow. The figure abstracts the common evaluation structure used for CGFormer and FoundationSSC variants: an image-context pathway and a geometry-depth pathway are modified independently while the downstream SSC pipeline is kept as close as possible to the original published architecture.

SSC designs whose image-context and geometry pathways are exposed enough to support controlled feature-transfer studies.

CGFormer and FoundationSSC are selected consciously even though their published forms use stereo-derived geometry priors rather than the monocular MDE or pose-aware MVS depth estimators targeted in this chapter. The reason is methodological: both expose a useful separation between image-context information and geometry information. CGFormer combines an image encoder, depth-context components including depth-aware view transformation, CGVT, a 3D local-global encoder, and a decoding head [40]. FoundationSSC is a dual-decoupling framework with a foundation encoder, decoupled semantic-context and geometric pathways, hybrid view transformation, axis-aware fusion, and a decoding head [6]. We adapt these published designs for controlled evaluation.

CGFormer Variant: Frozen Depth Prior Substitution The modification is to use precomputed depth maps from frozen pretrained depth heads as replacement geometry priors for CGFormer’s original stereo-depth or depth-map input slot. The depth-prior sources considered are from Depth Anything V2 fine tuned on Occufly (cite occufly dataset) and V-JEPA 2.1 depth head DPT-FiLM trained on Occufly. SSC metrics like occupied mean intersection over union (mIoU) for measure of sematic class matching and intersection over union (IoU) for measure of occupied occupy are reported as a preliminary comparison.

FoundationSSC Variant FoundationSSC is used as a stronger published follow-up baseline because it explicitly decouples semantic-context and geometric information at both the source and pathway levels [6]. In its published implementation, a frozen FoundationStereo encoder provides image-context features, cost-volume geometry features, and a dense depth map. Compared with CG-

Table 3: SSC variants and the geometric prior isolated by each design.

Variant	Geometric prior	Question isolated
CGFormer compar- ison	Exported monocular pri- ors	Do frozen V-JEPA 2.1-derived depth priors change a fixed CGFormer-style SSC evaluation path?

Former, FoundationSSC is reported in the published comparison to improve on
 SemanticKITTI and SSCBench-KITTI-360, with the gains attributed to the de-
 coupled foundation encoder-pathways, learned disparity-to-depth volume map-
 ping, hybrid view transformation, and axis-aware fusion [6, 40]. We therefore use
 it as a suitable architecture for ablation studies.

Considered Losses

Evaluation Metrics SSC evaluation uses IoU and mIoU, following the pub-
 lished CGFormer and FoundationSSC evaluation protocol [6, 40]. IoU is the ge-
 ometric occupied-voxel completion metric: all non-empty labels are treated as
 occupied, and the score measures occupied-vs-empty completion. mIoU is the
 voxel-wise semantic completion metric averaged over semantic classes.

4 Experimental Setup and Results

4.1 Dataset and Tasks

We evaluate on OccuFly [9] for monocular metric depth estimation and seman-
 tic scene completion (SSC). For the standard protocol, we use scenes 01–05 for
 training, scenes 06–07 for validation, and scenes 08–09 for testing, across alti-
 tudes of 30, 40, and 50 m. The standard depth split contains 14,804 training
 frames, 1,965 validation frames, and 3,842 test frames. RGB inputs are resized
 to 384×384 . Depth values are represented in meters, clipped to $[0.001, 80]$, and
 evaluated only over finite positive pixels that remain valid after mask-aware
 resizing.

For SSC, OccuFly labels are defined on a voxel grid of shape $192 \times 128 \times 128$
 in WHD order. The grid spans $x \in [-48, 48]$ m, $y \in [-32, 32]$ m, and $z \in [0, 64]$
 m, corresponding to a voxel size of 0.5 m.

4.2 MDE benchmarking results

All experiments are run on an NVIDIA DGX node with 8 NVIDIA H100 80GB
 GPUs.

Table 4: OccuFly test monocular depth estimation comparison. Published baseline values for MapAnythingV1.1, Metric3Dv2, Depth Anything 2, and Depth Anything 3 follow Table 5 of OccuFly [9]. Light pink marks the best RMSE.

Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
Linear Probe RGB	0.651	0.958	0.996	0.189	7.808	6.457	0.226
Linear Probe Video 4 frame clip	0.692	0.953	0.994	0.190	7.226	5.996	0.220
SFP_FiLM_DPT	0.808	0.953	0.993	0.143	5.160	3.732	0.189
ML_DPT	0.469	0.795	0.898	0.267	11.471	9.227	0.411
ML_FiLM_DPT RGB	0.833	0.971	0.998	0.129	4.621	3.523	0.164
ML_FiLM_DPT (DTA) Video	0.805	0.970	0.998	0.136	4.878	3.635	0.167
DepthAnything2_OccuFly [9]	0.834	0.976	0.999	0.134	4.844	4.193	0.112
Metric3Dv2_OccuFly [9]	0.278	0.806	0.985	0.381	13.643	13.134	0.098
MapAnythingV1.1 [15]	0.000	0.000	0.003	0.799	30.068	29.309	0.069
Metric3Dv2 [12]	0.073	0.208	0.455	0.471	19.578	18.409	0.156
DepthAnything2 [37]	0.002	0.015	0.059	0.729	28.382	27.392	0.192
DepthAnything3 [19]	0.000	0.005	0.141	0.591	22.615	22.019	0.043

Table 5: Best depth result for the altitude-interpolation split: training uses all scenes 01–09 at 30 m and 50 m altitude, validation uses scenes 01–04 at 40 m altitude, and testing uses scenes 05–08 at 40 m altitude.

Backbone & Depth head	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
V-JEPA 2.1 RGB ML_FiLM_DPT	0.9653	0.9997	0.9999	0.1021	4.7071	3.9570	0.119599

4.3 SSC benchmarking results

Table 8 reports semantic scene completion performance on OccuFly. The published comparison rows reproduce the altitude-wise and overall values from OccuFly Table 4 [9] for Symphonies [14] and DISC [21]. The CGFormer and FoundationSSC rows report values exported by the local SSC benchmark dashboard.

5 Discussion and Conclusion

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Table 6: OccuFly test MDE comparison grouped by altitude, with Metric3Dv2-OccuFly and DepthAnything2-OccuFly altitude-wise values appended from OccuFly Table 12 [9]. Light pink marks the best RMSE within each altitude group.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
30	Linear Probe RGB	0.743	0.952	0.986	0.205	6.280	5.391	17.585
	Linear Probe Video 4 frame clip	0.512	0.933	0.979	0.266	7.942	7.002	17.676
	SFP_FiLM_DPT	0.889	0.961	0.991	0.111	4.022	2.847	16.638
	ML_DPT	0.415	0.889	0.940	0.295	9.505	8.238	35.043
	ML_FiLM_DPT RGB	0.917	0.980	0.999	0.097	3.448	2.571	13.185
	ML_FiLM_DPT (DTA) Video	0.944	0.982	0.999	0.081	2.939	2.051	12.356
	Metric3Dv2-OccuFly [9]	0.460	0.757	0.992	0.347	10.917	10.215	0.101
	DepthAnything2-OccuFly [9]	0.919	0.982	0.998	0.103	3.108	2.666	0.086
40	Linear Probe RGB	0.730	0.943	0.998	0.180	5.851	4.804	21.671
	Linear Probe Video 4 frame clip	0.720	0.924	0.996	0.185	5.763	4.510	20.695
	SFP_FiLM_DPT	0.729	0.911	0.988	0.186	5.665	4.056	21.991
	ML_DPT	0.577	0.794	0.905	0.239	9.106	6.916	39.112
	ML_FiLM_DPT RGB	0.761	0.945	0.995	0.158	4.804	3.588	18.577
	ML_FiLM_DPT (DTA) Video	0.737	0.938	0.997	0.173	5.191	3.800	19.481
	Metric3Dv2-OccuFly [9]	0.209	0.802	0.981	0.395	13.138	12.625	0.096
	DepthAnything2-OccuFly [9]	0.795	0.957	0.998	0.148	4.486	3.823	0.119
50	Linear Probe RGB	0.524	0.978	1.000	0.192	9.981	8.728	17.395
	Linear Probe Video 4 frame clip	0.741	0.994	1.000	0.161	8.244	7.158	16.681
	SFP_FiLM_DPT	0.858	0.995	1.000	0.110	5.038	3.777	13.753
	ML_DPT	0.375	0.754	0.872	0.285	14.243	12.179	37.819
	ML_FiLM_DPT RGB	0.873	0.995	1.000	0.113	4.872	3.879	12.947
	ML_FiLM_DPT (DTA) Video	0.817	0.998	1.000	0.122	5.213	4.165	13.538
	Metric3Dv2-OccuFly [9]	0.283	0.831	0.987	0.379	15.373	14.950	0.099
	DepthAnything2-OccuFly [9]	0.844	0.997	1.000	0.129	5.985	5.261	0.114

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Table 7: OccuFly test MDE comparison grouped by scene.

Group	Method	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$	AbsRel $\downarrow$	RMSE $\downarrow$	MAE $\downarrow$	SILog $\downarrow$
scene_08	Linear Probe RGB	0.752	0.985	0.999	0.159	6.972	5.715	18.664
	Linear Probe Video 4 frame clip	0.665	0.983	0.999	0.183	7.465	6.358	19.247
	SFP_FiLM_DPT	0.976	0.998	1.000	0.069	3.014	2.468	7.813
	ML_DPT	0.601	0.946	0.976	0.212	8.949	7.353	27.068
	ML_FiLM_DPT RGB	0.975	0.997	1.000	0.067	2.934	2.392	7.869
	ML_FiLM_DPT (DTA) Video	0.982	0.998	1.000	0.055	2.566	2.010	7.090
scene_09	Linear Probe RGB	0.610	0.947	0.995	0.201	8.121	6.757	23.551
	Linear Probe Video 4 frame clip	0.702	0.941	0.993	0.192	7.129	5.851	22.872
	SFP_FiLM_DPT	0.741	0.935	0.990	0.173	5.805	4.242	20.643
	ML_DPT	0.415	0.734	0.866	0.289	12.343	9.983	42.988
	ML_FiLM_DPT RGB	0.776	0.960	0.997	0.155	5.147	3.979	17.416
	ML_FiLM_DPT (DTA) Video	0.734	0.958	0.998	0.169	5.542	4.287	17.872

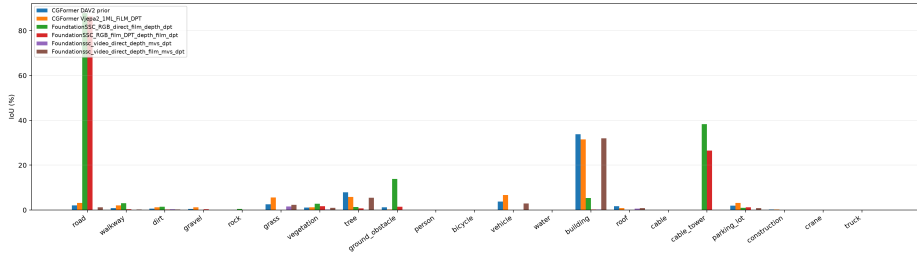


Fig. 3: OccuFly SSC per-class IoU bar chart for the evaluated CGFormer and FoundationSSC variants.

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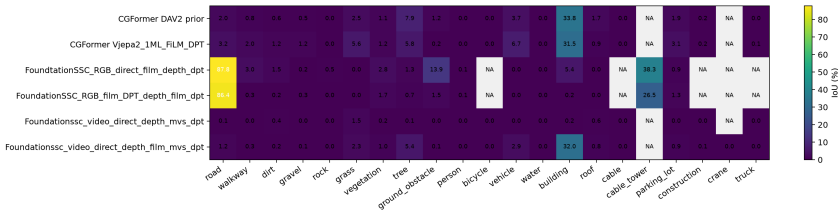


Fig. 4: OccuFly SSC per-class IoU heatmap for the evaluated CGFormer and FoundationSSC variants.

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Table 8: OccuFly SSC benchmark comparison by altitude. Published Symphonies and DISC values follow OccuFly Table 4 [9]; CGFormer and FoundationSSC values are from the local SSC benchmark report. For our variants, light beige marks the best SC IoU and light pink marks the best SSC mIoU within each altitude group and the overall All group.

Altitude Method		SC IoU $\uparrow$	SSC mIoU $\uparrow$
<i>Published OccuFly Table 4 baselines</i>			
50 m	Symphonies [14]	15.88	0.58
50 m	DISC [21]	31.10	2.20
40 m	Symphonies [14]	10.71	0.52
40 m	DISC [21]	27.85	1.77
30 m	Symphonies [14]	13.22	0.76
30 m	DISC [21]	26.88	2.23
All	Symphonies [14]	13.68	0.58
All	DISC [21]	29.52	2.04
<i>Our CGFormer and FoundationSSC variants</i>			
30 m	CGFormer DAV2 prior	36.10	4.22
30 m	CGFormer V-JEPA 2.1 ML_FiLM_DPT prior	36.70	4.39
30 m	FoundationSSC RGB direct FiLM depth DPT	33.21	5.16
30 m	FoundationSSC RGB FiLM-DPT depth FiLM-DPT	6.74	0.26
30 m	FoundationSSC video direct depth MVS DPT	2.84	0.18
30 m	FoundationSSC video direct depth FiLM-MVS-DPT	35.59	5.06
40 m	CGFormer DAV2 prior	38.42	2.86
40 m	CGFormer V-JEPA 2.1 ML_FiLM_DPT prior	29.05	2.12
40 m	FoundationSSC RGB direct FiLM depth DPT	27.85	4.08
40 m	FoundationSSC RGB FiLM-DPT depth FiLM-DPT	22.15	1.69
40 m	FoundationSSC video direct depth MVS DPT	12.30	0.87
40 m	FoundationSSC video direct depth FiLM-MVS-DPT	29.22	3.97
50 m	CGFormer DAV2 prior	35.56	2.81
50 m	CGFormer V-JEPA 2.1 ML_FiLM_DPT prior	42.78	4.06
50 m	FoundationSSC RGB direct FiLM depth DPT	21.16	2.53
50 m	FoundationSSC RGB FiLM-DPT depth FiLM-DPT	19.51	0.75
50 m	FoundationSSC video direct depth MVS DPT	5.21	0.21
50 m	FoundationSSC video direct depth FiLM-MVS-DPT	22.28	2.59
All	CGFormer DAV2 prior	36.77	3.05
All	CGFormer V-JEPA 2.1 ML_FiLM_DPT prior	36.51	3.31
All	FoundationSSC RGB direct FiLM depth DPT	37.68	3.46
All	FoundationSSC RGB FiLM-DPT depth FiLM-DPT	35.47	1.73
All	FoundationSSC video direct depth MVS DPT	24.55	0.17
All	FoundationSSC video direct depth FiLM-MVS-DPT	32.55	2.36